

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x . The product of **12** and the fourth odd integer is at most **26** less than the sum of the first and third odd integers. Which inequality represents this situation?

Answer

- A. $12(x + 6) \leq x + (x + 4) - 26$
- B. $12(x + 6) \geq 26 - (x + (x + 4))$
- C. $12(x + 4) \leq x + (x + 3) - 26$
- D. $12(x + 4) \geq 26 - (x + (x + 3))$

Correct Answer: A

Rationale

Choice A is correct. It’s given that the four odd integers are consecutive, ordered from least to greatest, and that the first odd integer is represented by x . It follows that the second odd integer is represented by $x + 2$, the third odd integer is represented by $x + 4$, and the fourth odd integer is represented by $x + 6$. Therefore, the product of **12** and the fourth odd integer is represented by $12(x + 6)$, and **26** less than the sum of the first and third odd integers is represented by $x + (x + 4) - 26$. Since the product of **12** and the fourth odd integer is at most **26** less than the sum of the first and third odd integers, it follows that $12(x + 6) \leq x + (x + 4) - 26$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.